

Using **Compute** to create a New Variable from a formula

Compute Variable

Transform → Compute Variable

- The name of the new variable goes in the “Target Variable” window
- Put the formula for the new variable in the “Numeric Expression” window — you can either type in the formula or use the point-and-click buttons
- The “Functions” list provides a variety of common transformations

Example: We want to compute “age_diff” as the difference between the actor playing Clark Kent/Superman and Lois Lane.

SPSS Syntax

```
COMPUTE age_diff = clark_age - lois_age.  
VARIABLE LABEL age_diff "age difference  
between Clark Kent and Lois Lane".  
EXE.
```

- ① Create new variable from a formula using existing variables
- ② Always add a descriptive variable label after computing a new variable

i Note

If a compute statement includes a variable with a missing value for a case, then the resulting variable also has a missing value for that case.

Using Recode to Combine Categories of a Qualitative Variable

Recode Into Different Variables

Transform → Recode → Into Different Variables

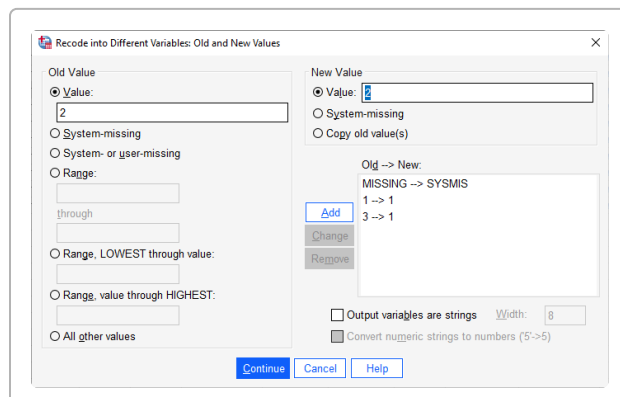
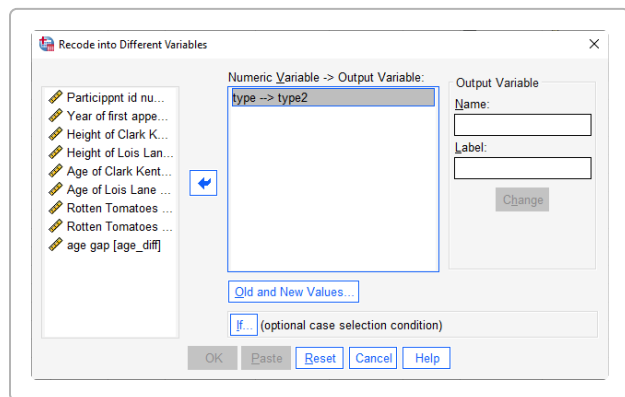
- Highlight the variable in the list and click the arrow to move it into the middle window
- Type the new variable name in the “Name” box under “Output Variable”
- Click the “Change” button
- Click the “Old and New Values” button to open the recoding window

Example: The “type” variable has three categories. We want to create a new variable “type2” that combines Film and Serial into one category and keeps TV Series as a second category.

SPSS Syntax

```
RECODE type
  (MISSING=SYSMIS) (1=1) (3=1) (2=2)
  INTO type2.
VARIABLE LABEL type2 "Type of media recoded".
VALUE LABELS type2
  1 "Film & Serial"
  2 "TV Series".
EXE.
```

- 1 Specify the source variable and define old-to-new value mappings (handle missing first!)
- 2 Name the new output variable
- 3 Add variable label and value labels for the new variable



Using Recode to Make a Quantitative Variable into Ordered Categories

You can also use Recode to convert a quantitative variable into categories using ranges.

Example: We want to create a new variable “age_grp3” that categorizes the age difference between actors into three groups: small (under 5 years), moderate (5–10 years), and large (over 10 years).

Recode with Ranges

Transform → Recode → Into Different Variables

- Use the **Range** buttons on the “Old Value” side to specify what range of original values get recoded into the new value
 - “Range, value through highest:” for values above a threshold
 - “Range, Lowest through value:” for values below a threshold
 - “Range:” for values between two thresholds
- On the “New Value” side, enter the new category code
- Click “Add” after each mapping

SPSS Syntax

```
RECODE age_diff
  (MISSING=SYSMIS)
  (Lowest thru 4.99=1) (5 thru 10=2) (10.01 thru Highest=3)
  INTO age_grp3.
VARIABLE LABEL age_grp3 "Age difference category".
VALUE LABELS age_grp3
  1 "Small (< 5 yrs)"
  2 "Moderate (5-10 yrs)"
  3 "Large (> 10 yrs)".
EXE.
```

- ① Always handle missing values first — map them to system missing
- ② Use **Lowest thru** and **thru Highest** for open-ended ranges
- ③ Name the new output variable
- ④ Add variable label and value labels

Taking Care of Missing Values in Recode

! Important

When recoding, you must tell SPSS what to do with missing values, or they may be left undefined in your new variable!

- On the **Old Value** side — click the radio button beside “System- or user-missing”
- On the **New Value** side — click the radio button beside “System-missing”
- Click “Add”

Do this **before** specifying your other recode mappings.

Using the SPSS Syntax Window

SPSS Syntax Editor

File → New → Syntax

- Type or paste SPSS commands into the Syntax Editor window
- To run a command, highlight it and click the green arrow button on the toolbar
- You can run all commands at once, or highlight and run them one at a time

Syntax Conventions

- **CAPITAL LETTERS** are used for SPSS commands and keywords (`COMPUTE` , `RECODE` , `IF` , `EXE`). Lowercase is used for variable names and values. SPSS is not case-sensitive, but this convention makes syntax easier to read.
- A **period** (`.`) marks the end of a command. Every command must end with a period.
- A **forward slash** (`/`) separates sub-commands within a command (e.g., `/STATISTICS` , `/HISTOGRAM`).

! Important

Understanding the `EXE.` command: When you run a transformation (Compute or Recode), SPSS creates the new variable but the values are not actually computed until either (1) you request any statistical analysis, or (2) you include and run an `EXE.` (short for `EXECUTE.`) command. It is good practice to include `EXE.` after your transformations so the values are computed immediately.